

CYRIL JOHN

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U.S. Citizen

EDUCATION

Texas A&M University

Dec 2024

BSc Computer Science, Minor in Business

GPA 3.43

- **Relevant Coursework:** Discrete Math, Data Structures & Algorithms, Parallel Computing, OOP, Statistics, Database
- **Honor:** Dean's List 5 semesters, Senior Capstone Best App UI, Best Pitch at McFerrin Startup Competition

EXPERIENCE

Texas A&M Brain Networks Lab | Research Assistant

Aug - Dec 2024

- Improved a reinforcement learning framework with **PyTorch** to train DQN agents and cut training time 8x on **CUDA**
- Implemented custom loss functions with **NumPy** and **SciPy**, improving model stability and cutting crashes by 50%
- Enhanced experiment visualization using **Matplotlib**, **OpenCV** and **Jupyter** to generate live metrics on 100+ episodes

Honeywell | Software Engineer

May - Aug 2023

- Built a data automation web app with **Node.js** and **React** that streamlined the preprocessing of 10mb+ CSV files from 1 hour to under 10 minutes for the Dynamo Alarm Management team
- Utilized React **Hooks** for stateful UI, **Axios** for API calls and **Material-UI** components for dynamic table rendering
- Developed 5 **REST** endpoints in **Express.js** with **PapaParse**, validated functionality with **Postman** unit tests

PROJECTS

Capstone Matching iOS App (Flask, Supabase, Python, SQLAlchemy) | [GitHub](#)

- Developed a **Python** recommendation engine with decision tree logic to connect users with hairstylists
- Built 7+ **REST APIs** in **Flask** with **SQLAlchemy** models for clients, hair profiles, ratings and liking functionalities
- Designed 7+ relational models in **Supabase Postgres** and implemented authentication for secure email token login
- Integrated **Google Maps API** to geocode addresses and store latitude/longitude enabling radius-based filtering

Premier League Odds (Angular, FastAPI, PyTorch, pandas, AWS, SQLite) | [GitHub](#)

- Built a **FastAPI** and **Angular** web app integrating logistic regression in **PyTorch** to predict Premier League games
- Collected and processed 10 seasons of match data from API-Football with **Python** request, **pandas** and **SQLite**
- Deployed application on AWS using **Lambda**, **API Gateway**, and **S3** with caching to minimize API usage

Restaurant POS & Ordering App (Django, PostgreSQL, AWS, Docker) | [GitHub](#)

- Designed and implemented a relational schema in **PostgreSQL** to support restaurant management and food ordering
- Built a Django user model and developed authentication flows in **Django** using **OAuth2** with 5 **REST API** endpoints
- Containerized the backend and database with **Docker** and deployed on AWS, services on **EC2** and static assets on **S3**

AI Docstring Generator (Python, LibCST, Typer, Ollama, LLaMA-3) | [GitHub](#)

- Developed a **CLI** tool to scan Python codebases and auto-generate Google-style docstrings using **LLaMa-3** and **LibCST**
- Implemented prompt engineering with function signatures and validation guards in a **Typer** based workflow
- Augmented with a **RAG** pipeline using **FAISS** and local embeddings to add context from code, tests and docs

Emotion Recognition System (OpenCV, Mediapipe, scikit-learn) | [GitHub](#)

- Built a real-time facial emotion recognition pipeline using **OpenCV** to capture frames from a live desktop camera and **Mediapipe FaceMesh** to extract 1400+ facial landmarks per frame
- Engineered a **Random Forest classifier** in **scikit-learn** to categorize emotions achieving 80% accuracy on test splits

LEADERSHIP

Engineers Without Borders | Project Manager

- Directed a team of **50** students and facilitated collaboration with community leaders, NGOs, and INTEC to address open ditches in low-lying areas of a Dominican Republic community through new bridge construction

SKILLS

- **Frontend:** JavaScript/TypeScript React, Next.js Angular, Framer Motion
- **Backend:** Python, Django, Flask, FastAPI, SQLAlchemy, Node.js, PyTorch, NumPy, pandas, TensorFlow, LLM, RAG
- **Database and Cloud:** AWS CCP, AWS (Lambda, API Gateway, EC2, S3), Vercel, Docker, Git, Postman, PostgreSQL